Capstone Design

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Verified Study Buddy

Capstone Design

> Prepare Capstone Design: check the target, prerequisites, resources and adaptable teaching schedule.

Teach the learning cycle

Primary target: Explain the **mechanism** and terminology of capstone **requirement**s and corpus

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min

> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use “Run the investigation”, “Connect your project”, “Show what you know”, “Study a complete case” and “Try a new case independently”. Give extra time for the full task set. If this week has a coding lab, use its named section; section numbers can change when extra explanations are included.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Case exposure check: this source example appears in ai-3/19, ai-3/33. Reuse a familiar case for recall or modelling; verify that the case used to assess transfer has not already received feedback.

Retrieval answer guidance

Human AI Workflow

Without opening the old lesson, explain one key idea from Human AI Workflow. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of human review and consequential use

Reference example, not the only acceptable wording: Teacher explanation: Human review needs a responsible actor, **evidence** packet and explicit authority. Adding a person to a workflow does not guarantee effective oversight.

For this fictional workflow the assistant drafts a room request but only the coordinator can approve and submit it. Review packets include sources, uncertainty, requested action and the exact version.

Any material change requires new approval.

Bias and Representation

Without opening the old lesson, explain one key idea from Bias and Representation. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of representation and bias evaluation

Reference example, not the only acceptable wording: Teacher explanation: Bias evaluation tests defined outcomes across relevant conditions.

Group averages can expose differences but cannot alone establish their cause or the correct remedy. Define the task, relevant groups, labels and denominators before comparing. Use fictional, non-identifying records.

Evaluate quality and fallback outcomes separately; preserve sample sizes and unresolved explanations.

Red Team the Product

Without opening the old lesson, explain one key idea from Red Team the Product. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of red team exercises

Reference example, not the only acceptable wording: Teacher explanation: Red-team exercises probe explicit failure boundaries in an authorised isolated fixture environment. Their purpose is **evidence**-based improvement.

Use fictional records and allowed test routes. Define a target contract violation, record the exact case, observe action/data traces and preserve a reproducible report. Do not touch external systems.

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

Design a chapter assistant with one answerable, one ambiguous and one unanswerable query. State the expected behaviour and one mandatory quality criterion.

Concept to explain

Requirements, **success measure**s, sources, risks, and tests precede building.

Term	Meaning
requirement	A behaviour or condition the proposed system must satisfy.
success measure	An observable quantity or criterion used to decide whether a task was achieved.

Term	Meaning
risk register	A record of identified risks, their consequences, mitigations and responsible reviewers.
design review	An examination of a proposed system against its requirements before proceeding.

Prepare the class

Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.

Alternative compact allocation

> This compact option is not the full-week schedule. Use the main teaching cycle and allow extra sessions for the investigation, independent work and project.

Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

Two 45-minute sessions. A grounded-assistant capstone begins with bounded **requirement**s and a versioned permitted corpus. A baseline can be a manual lookup **process** using the same **evidence**.

Use the supplied constructed scenarios to make the **mechanism** inspectable. Collect W4 before sharing solutions.

Students should name intermediate **evidence** and limitations; classroom pacing and learning effectiveness remain unvalidated.

Readiness answer

A result depends on the **input**s, procedure and conditions that produced it. Preserve data roles and distinguish observed outcomes from **prediction**s or unmeasured conditions.

If the prior concept cannot be explained, use a short retrieval activity before this chapter.

Lesson and workbook; paper and pencil

Calculator and optional local Python 3 for arithmetic; no live API or private data

Prepare the fresh retry separately and withhold its solution until after collection

End goal for the pictorial case

By the end, I can explain capstone design using a traced example and a stated limitation.

This added worked case illustrates the week’s main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

TEACHER GUIDE · 2 OF 9

Explain the mechanism

> Model Capstone Design using the supplied case, labelled mechanism and reasoning. Make the scope of each conclusion explicit.

Input → Process → Output

Trace the information. At each stage, ask what changed and how you could test it.

Which stage could fail even when the other two are correct? Explain:

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.

1. Need: answer questions from one approved science unit.

2. **Requirement**s include citations, age-fit explanation, and missing-source abstention.

3. Success: at least 8/10 fixed eval cases pass, with no unsupported claims.

4. Review architecture, privacy, provenance, and safety before building.

Recalculate, retrace, or restate the reasoning in your own representation:

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

Concrete teaching case

The teaching case

Capstone: answer from one approved chapter with citations and a missing-**evidence** fallback.

The teaching case

Define acceptance tests.

Exact explanation to model

Why the result follows

Use answerable, ambiguous and unanswerable cases. Check correctness, support, format and fallback, plus declared permissions and privacy.

Explain the added worked example

A grounded-assistant capstone begins with bounded **requirement**s and a versioned permitted corpus. A baseline can be a manual lookup **process** using the same **evidence**.

Explain the added worked example · 1

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

Specify the user question, source authority, access, **output**/fallback contract and evaluation plan. Freeze corpus IDs and split cases before implementation. Keep teacher material out of student retrieval.

Explain the added worked example · 2

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Key terms

Term	Meaning
Requirement	A declared user-visible condition the system must satisfy
Corpus manifest	An inventory of source IDs, versions and access roles
Baseline procedure	A credible reference process on the same task

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Student room assistant corpus contains public register P, teacher-only guide T and current approval A. It must report approved room capacity or needs_review.

Authorised corpus contains P and A, excluding T. Baseline manually checks same-room current approval and capacity. Reserve complete, missing and conflicting cases before development.

Model the reasoning aloud · 3

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Capstone Design

Plan a bounded study buddy around a small approved vocabulary corpus.

The facts we will use

> **Supplied facts**
>
> Corpus V1 defines roots, stem and leaf. It contains no plant disease treatment. Success requires a source reference; missing definitions require a clear fallback.

Supplied facts

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-3/33

Capstone Design

Start with the supplied facts
Follow each arrow. Ask which fact allows the next step.

1. Define contract

Answer only glossary-definition requests supported by V1.

2. Plan normal test

"What is a stem?" must point to V1's stem entry.

3. Plan missing test

"How do I cure this disease?" must produce an out-of-scope response, not a treatment guess.

CHANGE ONE THING
The corpus gains a teacher-approved flower definition in V2.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Define contract	Answer only glossary-definition requests supported by V1.
2. Plan normal test	“What is a stem?” must point to V1’s stem entry.
3. Plan missing test	“How do I cure this disease?” must produce an out-of-scope response, not a treatment guess.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

Read the picture in order. “How do I cure this disease?” must produce an out-of-scope response, not a treatment guess. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

What this example does not establish

Keep the stated limit

A capstone design is a testable plan; its proposed behaviours are not verified outcomes until checked.

Change one thing in the pictured case

Change one thing

Change: The corpus gains a teacher-approved flower definition in V2.

Predicted result

Prediction: Add a flower test using V2 and record the corpus version.

Why the result follows

Reason: The **evidence** boundary changed explicitly; the earlier corpus could not support this new entry.

Ask while showing the picture

Ask while showing the picture · 4

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

Demonstrate and guide practice

> Demonstrate Capstone Design, reduce support deliberately and check individual reasoning before discussing answers.

Board sequence

1. Put the facts on the board

Write the given facts: Capstone: answer from one approved chapter with citations and a missing-**evidence** fallback.

2. State the decision to explain

Write the question to explain: Define acceptance tests.

3. Reveal the worked reasoning

Reveal after **prediction**s: Use answerable, ambiguous and unanswerable cases. Check correctness, support, format and fallback, plus declared permissions and privacy.

Hold back the independent task

Keep for an independent attempt: The capstone goal is helps students. Rewrite it as an observable test for a passage-based study assistant.

Classroom dialogue

Ask the class

Ask: Which facts are given, and which would we have to assume? Learners mark relevant

details before discussing answers.

Language to model

Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

In pairs, one learner solves while the other checks **evidence**; exchange roles. Remove the worked answer before independent assessment.

Model these guided questions

Guided: Authorised corpus contains P and A, excluding T. Baseline manually checks same-room current approval and capacity. Reserve complete, missing and conflicting cases before development.

Qualification: A complete capstone plan is not **evidence** that the implementation meets it; evaluate after building.

Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the **mechanism**; reject a claim resting on unmeasured **evidence**.

Case-specific teaching sequence

Stage	Teacher action
Session 1: 0–7 minutes	Retrieve the prerequisite; identify the input, procedure and evidence source.
7–18 minutes	Explain section 1 and vocabulary. Complete W1 with a mechanism-based explanation.
18–32 minutes	Trace the declared procedure and worked case. Complete W2, recording intermediate steps.

Stage	Teacher action
32–45 minutes	Discuss the diagram with the worked input; use W3 to distinguish measured claims and limitations.
Session 2: 0–8 minutes	Retrieve the worked reasoning and collect the guided response before discussing it.
8–20 minutes	Compare a proposed boundary variant with the original case; identify what remains fixed.
20–35 minutes	Collect W4 independently. Require the procedure trace and the bounded conclusion.
35–45 minutes	Use W5 and exit responses for feedback. Administer the fresh retry after feedback and preserve both attempts.

Reduce help deliberately

Completion practice

Completion task: Student room assistant corpus contains public register P, teacher-only guide T and current approval A. It must report approved room capacity or needs_review.

Show the intermediate steps and give the conclusion with its scope.

Reasoning to listen for

Teacher check: Teacher explanation: Authorised corpus contains P and A, excluding T. Baseline manually checks same-room current approval and capacity. Reserve complete, missing and conflicting cases before development.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual reasoning check

Use the worked case for Capstone **requirement**s and corpus. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

> **Expected repair**
>
> Expected correction: A complete capstone plan is not **evidence** that the implementation meets it; evaluate after building.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictorial practice question

What should the project plan record before the new flower test? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Reasoned teacher answer

Record approved corpus V2, the expected sourced definition and which evaluation cases belong to that version. Changed-case check: Add a flower test using V2 and record the corpus version.

The **evidence** boundary changed explicitly; the earlier corpus could not support this new entry.

Changed-case reasoning: Add a flower test using V2 and record the corpus version. The **evidence** boundary changed explicitly; the earlier corpus could not support this new entry.

Collect each learner’s explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Run the weekly evidence lab

> Run the Capstone Design investigation and use the first unsupported step to choose a teaching response.

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.

Write the brief, architecture, source list, schema, rubric, eval set, risks, and approval gates; peer-review for testability.

> What did the hardest test teach you that the normal test did not?

Make the lab runnable

Use the concrete case as the shared normal case and the independent question as a second case. Each pair creates a boundary or missing-information case and predicts its result before testing.

Keep the expected result separate from an observed mistake.

Record	What to capture
Input or evidence	Exact data, instruction or claim
Expected result	Answer or safe fallback declared before the run
Observed result	Actual result, including errors
Explanation	Rule, calculation or evidence explaining the outcome
Next test	One justified change and a new prediction

Diagnose and reteach

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation.

Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

The worked result establishes every future case: A complete capstone plan is not **evidence** that the implementation meets it; evaluate after building.

A missing **input** may be replaced without recording the change: A replacement changes the **evidence** or contract. Record the change and its justification; do not silently turn unknown into a measured value.

Repeating an assessed case is new independent **evidence**: A replay checks consistency. An example used for feedback or revision cannot later be described as untouched final evaluation.

Match the support to the first error

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

TEACHER GUIDE · 5 OF 9

Assess and explain the answers

> Use the Capstone Design questions, teacher keys and criteria after the relevant attempts. Previously discussed cases are practice; protect the separately selected independent task.

Worked case answer

Teacher key · Worked case answer

Use answerable, ambiguous and unanswerable cases. Check correctness, support, format and fallback, plus declared permissions and privacy.

Independent question

Independent question · withhold the key

The capstone goal is helps students. Rewrite it as an observable test for a passage-based study assistant.

Expected independent answer

Teacher key · Expected independent answer

For example: given five questions answerable from a supplied passage, answer each with supporting passage **evidence**, and abstain on two questions the passage cannot answer. State expected **output**s before testing.

What counts as evidence

Criterion	0 points	1 point	2 points
Result	Incorrect or absent	Partly correct	Correct with necessary conditions
Reasoning	No supporting trace	Some steps	Clear mechanism, calculation or source support
Limit	Overclaims certainty	Vague limitation	Relevant uncertainty or missing fact
Transfer	Copies without adapting	Adapts with prompts	Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold.

Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the **evidence** that challenges this claim: “A broad goal is already a measurable acceptance criterion.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

> #### Teacher key · Original transfer answer
>
> Supply a direct fact query with a supporting passage; an unclear 'it' query requiring clarification; and an absent fact requiring fallback. Require every factual answer claim to be supported by its cited passage.
>
> Preserve the cases before implementation.

Explanations for the additional workbook examples

W1 Explain the mechanism

Workbook question · W1

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

Teacher reasoning · W1

Teacher explanation: A grounded-assistant capstone begins with bounded **requirement**s and a versioned permitted corpus. A baseline can be a manual lookup **process** using the same **evidence**.

Specify the user question, source authority, access, **output**/fallback contract and evaluation plan. Freeze corpus IDs and split cases before implementation. Keep teacher material out of student retrieval.

W2 Trace the worked case

Workbook question · W2

Student room assistant corpus contains public register P, teacher-only guide T and current approval A. It must report approved room capacity or needs_review. Show the intermediate steps and give the conclusion with its scope.

Teacher reasoning · W2

Teacher explanation: Authorised corpus contains P and A, excluding T. Baseline manually checks same-room current approval and capacity. Reserve complete, missing and conflicting cases before development.

W3 Audit the limit

Workbook question · W3

A report omits this qualification: A complete capstone plan is not **evidence** that the implementation meets it; evaluate after building.

Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

Teacher reasoning · W3

Teacher explanation: A complete capstone plan is not **evidence** that the implementation meets it; evaluate after building.

State the observed result separately from the unverified or out-of-scope claim; preserve the original **evidence** rather than inventing a missing fact.

W4 Apply a new case independently

Workbook question · W4

Student equipment assistant has public inventory I, restricted purchase ledger L and public booking notice B. Define eligible corpus, baseline and a missing-**evidence** test.

Teacher reasoning · W4

Teacher explanation: Use I and B only; exclude L. Baseline manually matches item availability to authorised booking **evidence**. Missing notice should trigger the specified review outcome.

W5 Design a revealing follow-up

Workbook question · W5

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

Teacher reasoning · W5

Teacher explanation: Accept a concrete variant of the worked **input** or a missing/conflicting/boundary condition, with an expectation derived from the declared procedure.

Require preserved settings, original **output**s and data roles. A complete capstone plan is not **evidence** that the implementation meets it; evaluate after building.

The new case should test this particular **mechanism** rather than merely restate the original result.

Assess independent transfer

Independent question · withhold the key

Selected case: A science assistant may use only these notes: evaporation is liquid becoming vapour; condensation is vapour becoming liquid.

Specify behavior for “Define evaporation”, “Explain it”, and “What is tomorrow’s weather?”.

Selected case answer

> **Teacher key · Selected case answer**
>
> Answer the definition using the supplied note and identify its source. Clarify the referent of “it”. Weather is absent from these notes: say the **evidence** is unavailable rather than inventing a forecast.
>
> Require support for every factual claim.

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

> **Teacher key · Selected case answer**
>
> Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.
>
> Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

Connect and extend learning

> Connect Capstone Design to Verified Study Buddy, adapt support and plan a later changed-case check.

Verified Study Buddy

Freeze the Study Buddy v1 design and acceptance tests before implementation.

Keep the decision, **input**, result, explanation and remaining limitation in the project **evidence** record. Retrieve this record next week rather than restarting the project.

Support and stretch

Provide one short approved passage; separate instruction, **input**, **output** and **evidence**. Stretch: introduce an unsupported claim or conflicting source and validate claim by claim with a fallback.

Exit reflection

Ask for one decision the learner can now justify and one situation requiring more **evidence** or human review. Confidence ratings are separate from correct independent performance.

Bridge to the next week

Next: Week 34: Capstone Build. Retrieve today's **evidence** and identify what carries forward and what is new.

Support and fresh reassessment

Provide an **input**/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace.

Extension: compare two concrete variants of this **mechanism** while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

W4: Use I and B only; exclude L. Baseline manually matches item availability to authorised booking **evidence**. Missing notice should trigger the specified review outcome. Require intermediate reasoning and the stated scope.

Fresh retry: Public timetable S and restricted staff notes N; student assistant reports meeting time. Expected reasoning: Use S only.

Manual timetable lookup is baseline; missing current timetable gives fallback rather than staff-note access.

If students need support, record it and reassess with another new case rather than awarding independence to a copied answer.

Next connection: Capstone grounded assistant. Preserve the current case record and retrieve its **evidence** boundary before moving on.

Plan the delayed return

Suggested return points in this level: ai-3/34, ai-3/36. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Project evidence from this case

For your Verified Study Buddy, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

TEACHER GUIDE · 7 OF 9

Prepare, teach and assess

> Use this preparation summary alongside the detailed Capstone Design lesson; it is a checklist rather than a replacement for the full sequence.

Teaching target

Teaching target: Explain the **mechanism** and terminology of capstone **requirement**s and corpus

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. No paid AI account is required for the paper reasoning activities.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
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> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

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> **Resources:** Use “Run the investigation”, “Connect your project”, “Show what you know”, “Study a complete case” and “Try a new case independently”. Give extra time for the full task set. If this week has a coding lab, use its named section; section numbers can change when extra explanations are included.

Before class, solve the supported practice: Student room assistant corpus contains public register P, teacher-only guide T and current approval A. It must report approved room capacity or needs_review.

Show the intermediate steps and give the conclusion with its scope.

Supported practice answer

Reasoned practice answer: Teacher explanation: Authorised corpus contains P and A, excluding T. Baseline manually checks same-room current approval and capacity.

Reserve complete, missing and conflicting cases before development.

Check every learner before discussion: Use the worked case for Capstone **requirement**s and corpus. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Repair the misconception

Misconception repair: A complete capstone plan is not **evidence** that the implementation meets it; evaluate after building.

Support without giving away the answer

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extend the reasoning

Extension: The corpus gains a teacher-approved flower definition in V2. Require a **prediction** and a reason before comparing. Expected result: Add a flower test using V2 and record the corpus version.

Reason: The **evidence** boundary changed explicitly; the earlier corpus could not support this new entry.

Protect the independent attempt

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint

Project checkpoint: Present and defend. Demonstrate the system, explain one failure and

justify the next improvement with **evidence**.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

TEACHER GUIDE · 8 OF 9

Use feedback, labs and repair paths

> Use feedback, earlier review and portfolios purposefully. Apply the coding routine only when this lesson has an executable lab.

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

For later coding lessons

> This week has no code block. Use this routine when a later lesson includes an executable lab.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

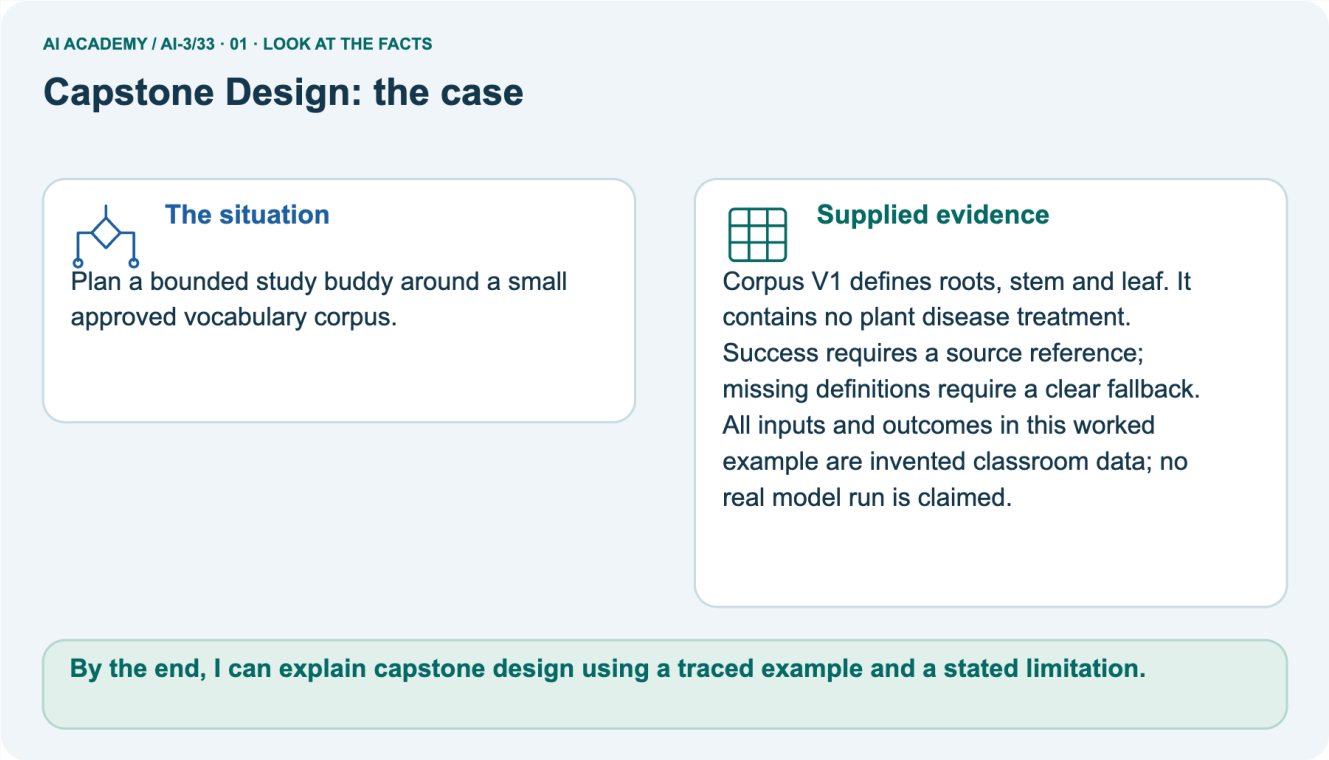
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Visual concept journey

> Show one Capstone Design graphic at a time, request a prediction and discuss its trace before the independent assessment.

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Capstone Design: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

Plan a bounded study buddy around a small approved vocabulary corpus. Supplied facts: Corpus V1 defines roots, stem and leaf. It contains no plant disease treatment.

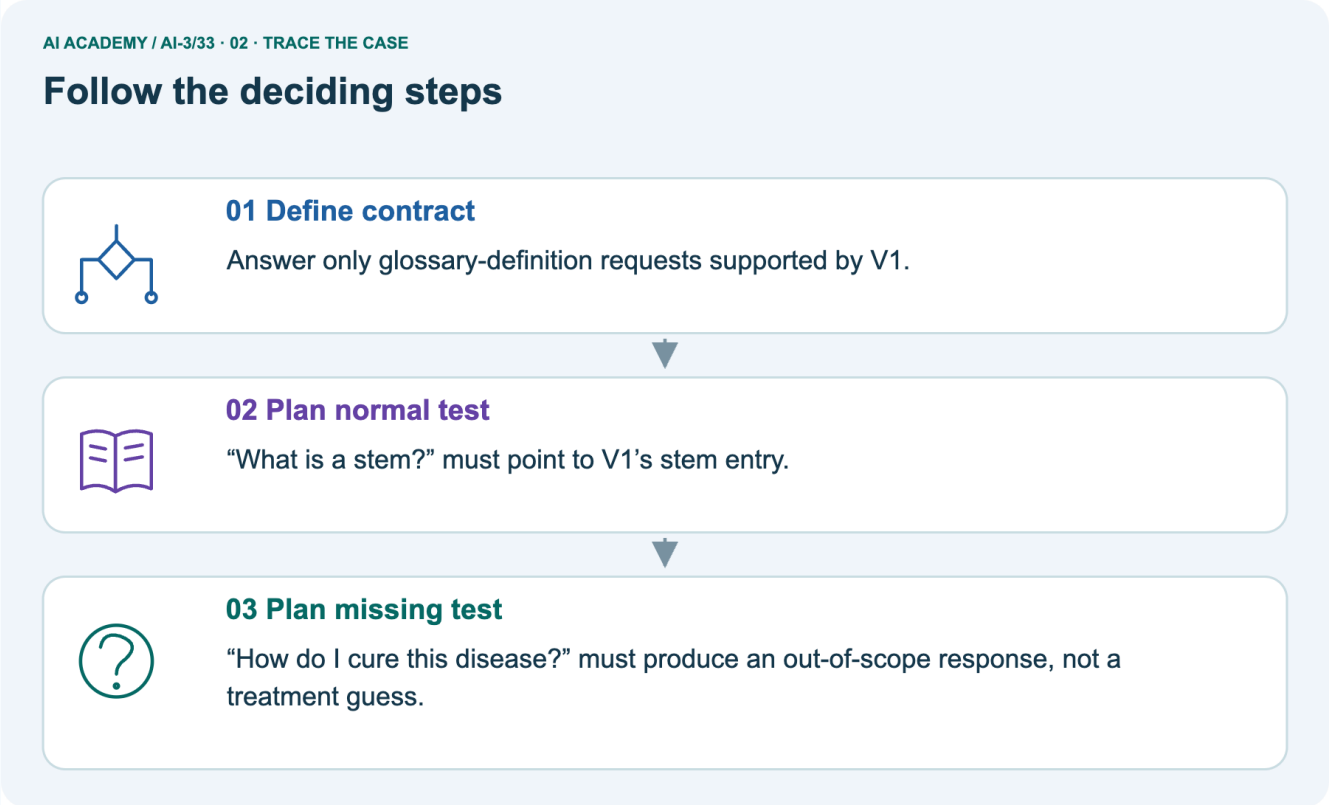
Success requires a source reference; missing definitions require a clear fallback. All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

> Point to the given fact that matters most. Explain why it matters.

Discuss while showing the picture

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

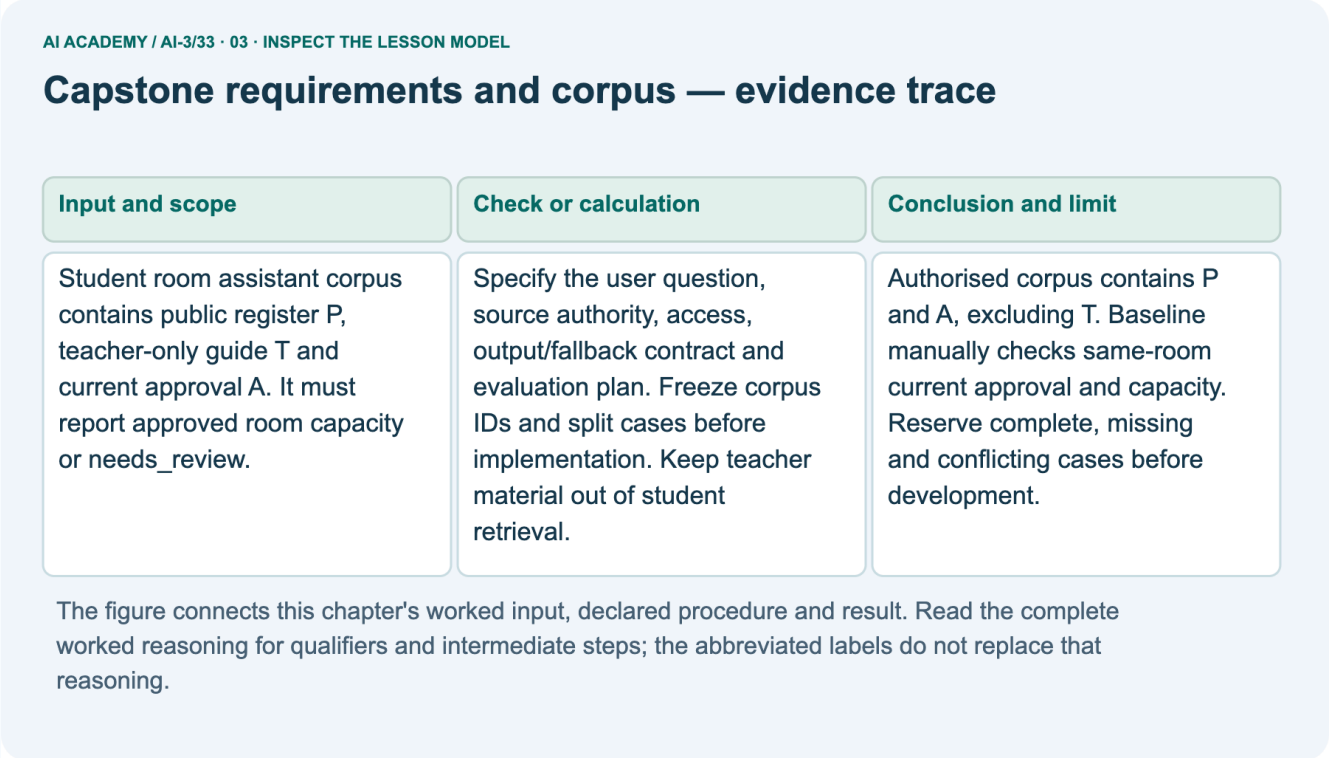
Define contract: Answer only glossary-definition requests supported by V1. → Plan normal test: “What is a stem?” must point to V1’s stem entry. → Plan missing test: “How do I cure this disease?” must produce an out-of-scope response, not a treatment guess.

Point to each numbered stage and explain which recorded **input** it uses.

Discuss while showing the picture

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Capstone requirements and corpus — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

Capstone **requirement**s and corpus — **evidence** trace. Student room assistant corpus contains public register P, teacher-only guide T and current approval A.

It must report approved room capacity or needs_review. | Specify the user question, source authority, access, **output**/fallback contract and evaluation plan. Freeze corpus IDs and split cases before implementation.

Keep teacher material out of student retrieval. | Authorised corpus contains P and A, excluding T. Baseline manually checks same-room current approval and capacity.

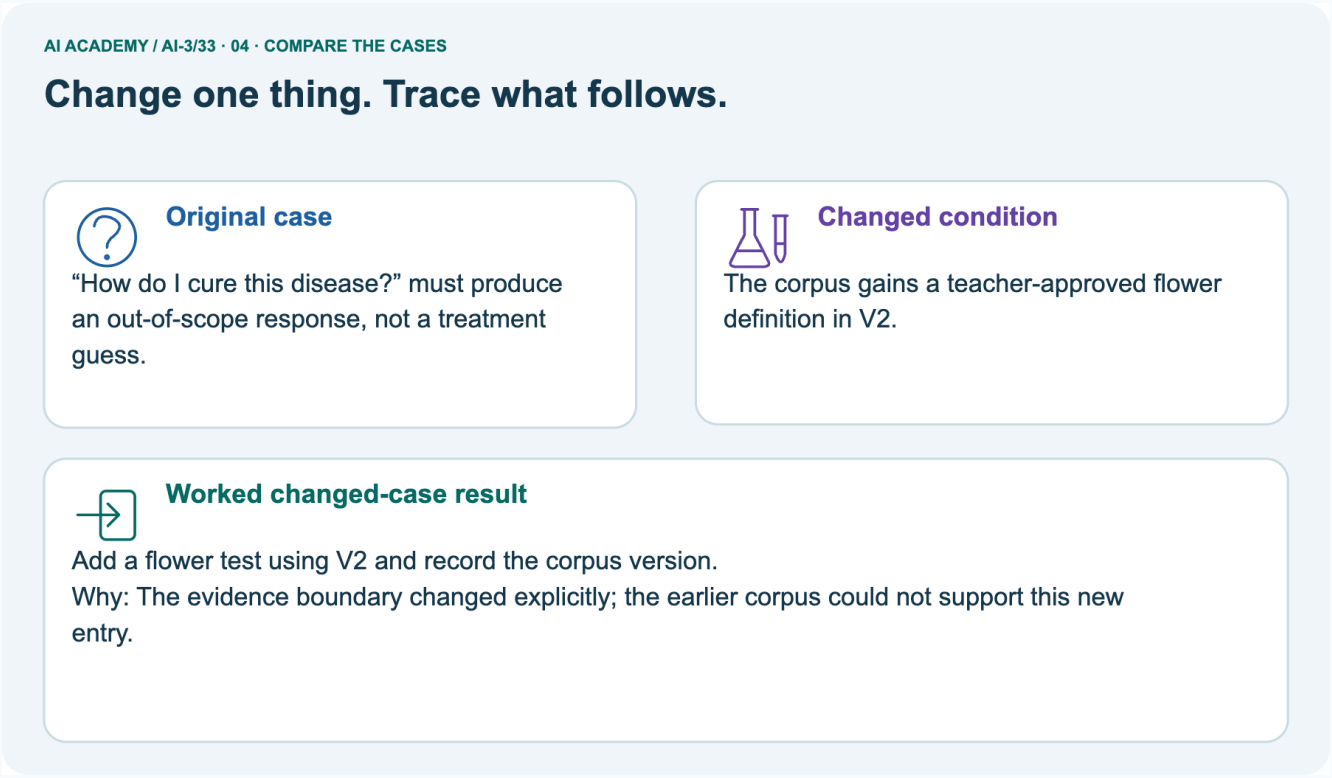
Reserve complete, missing and conflicting cases before development.

Read the labels and values. Explain the relationship the model shows.

Discuss while showing the picture

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: “How do I cure this disease?” must produce an out-of-scope response, not a treatment guess. Changed condition: The corpus gains a teacher-approved flower definition in V2.

Expected result: Add a flower test using V2 and record the corpus version. Reason: The **evidence** boundary changed explicitly; the earlier corpus could not support this new entry.

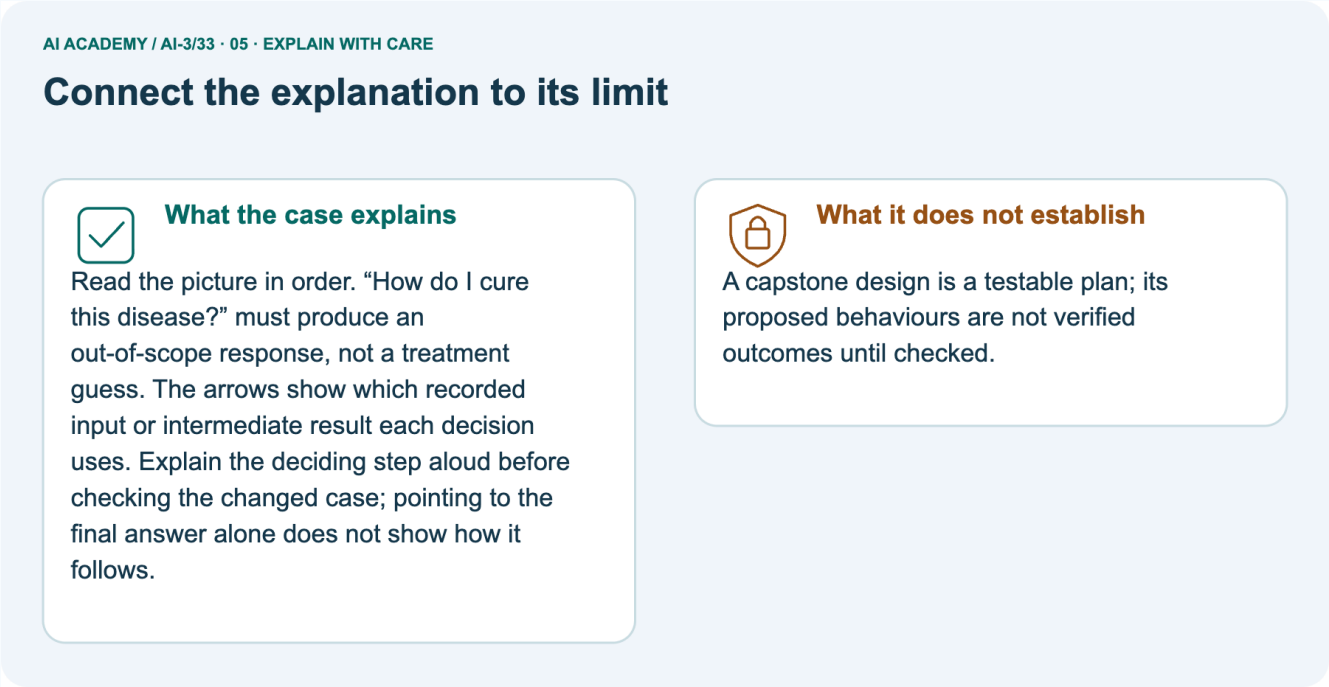
Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Discuss while showing the picture

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: The **evidence** boundary changed explicitly; the earlier corpus could not support this new entry.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture in order. “How do I cure this disease?” must produce an out-of-scope response, not a treatment guess. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Limit: A capstone design is a testable plan; its proposed behaviours are not verified outcomes until checked.

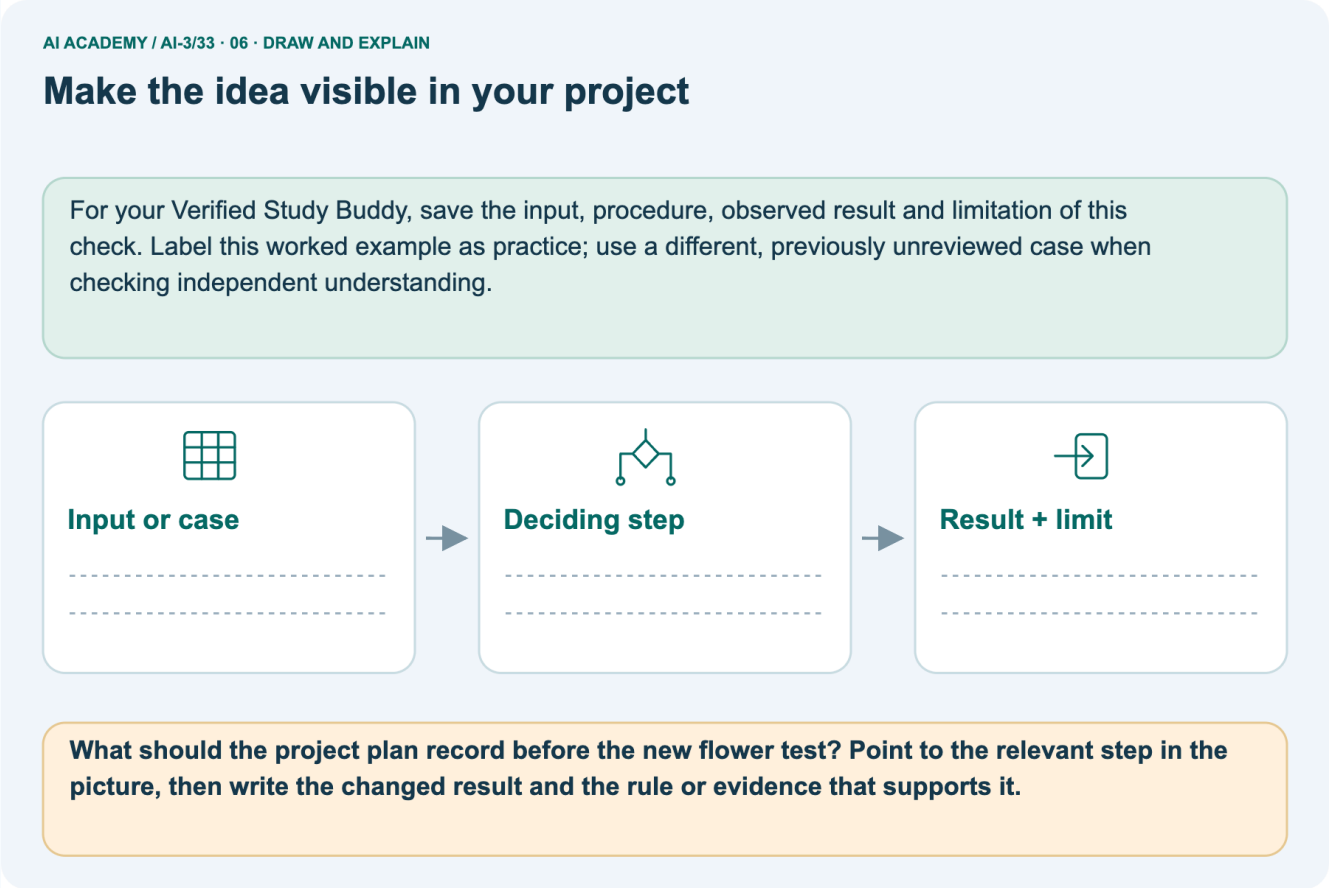
Explain the case, then point to a limit on the claim.

Discuss while showing the picture

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: A capstone design is a testable plan; its proposed behaviours are not verified outcomes until checked.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Verified Study Buddy, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: What should the project plan record before the new flower test? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

Discuss while showing the picture

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: Record approved corpus V2, the expected sourced definition and which evaluation cases belong to that version. Changed-case check: Add a flower test using V2 and record the corpus version.

The **evidence** boundary changed explicitly; the earlier corpus could not support this new entry.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.